Stationary Analysis of a Battery-Buffered EV Charging Station

A Piecewise-Deterministic Approach

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Outline

Model

Exponential Demands, $B = \infty$

Exponential Demands, $B < \infty$

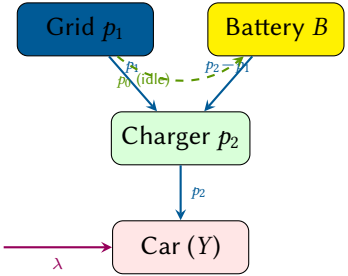
Performance Metrics

General Energy Demands

System Description

Physical setup

- One charging station, one charger
- Battery buffer of capacity $B \in (0, \infty]$
- Grid connection: idle charge rate p_0, service grid rate $p_1 > p_0$
- Fast charge rate $p_2 > p_1$ (grid + battery)
- Net battery discharge rate $c = p_2 - p_1$



Customers

- Cars arrive Poisson(λ)
- Energy demand $Y \sim F$, mean $1/\mu$
- **Loss system**: car arriving when busy is turned away

Two-Phase Service

Service mechanism

When a car is present, service proceeds in at most two phases:

Phase 1 — Fast charging ($x > 0$)

- Battery non-empty
- Grid + battery supply car at rate p_2
- Battery depletes: $\dot{x} = -c$

Phase 2 — Slow charging ($x = 0$)

- Battery empty
- Grid alone supplies car at rate $p_1 < p_2$
- Battery stays at 0: $\dot{x} = 0$

Key observation (memoryless demands)

If $Y \sim \mathrm{Exp}(\mu)$, residual demand at any time is again $\mathrm{Exp}(\mu)$. The departure rate is **Markovian**:

$$\delta(x) = \begin{cases} \mu p_2 & x > 0 \\ \mu p_1 & x = 0 \end{cases}$$

The PDMP Formulation

Definition

The process $Z(t) = (n(t), x(t))$ on $E = \{0, 1\} \times [0, B]$ is a **piecewise-deterministic Markov process** (Davis, 1984) with:

Flow $\dot{x} = r(n, x)$:

$$r(n,x) = \begin{cases} +p_0 & n=0,\ x<B \\ 0 & n=0,\ x=B \\ -c & n=1,\ x>0 \\ 0 & n=1,\ x=0 \end{cases}$$

Jump rate $\lambda(n, x)$:

$$\lambda(n,x) = \begin{cases} \lambda & n = 0 \\ \mu p_2 & n=1,\ x>0 \\ \mu p_1 & n=1,\ x=0 \end{cases}$$

Kernel Q:

$$Q\big((n,x),\cdot\big) = \delta_{(1-n,\,x)}$$

(only n flips; x unchanged at jumps)

Boundaries

- $x = 0$, $n = 1$: reflecting (slow-charging mode, δ switches to μp_1)
- $x = B$, $n = 0$: reflecting (battery full, $\dot{x} = 0$)

State Space and Stationary Distribution Structure

For $B = \infty$:

- State space $E = \{0, 1\} \times [0, \infty)$; **no upper boundary**
- No point mass $\pi_0^{(B)}$
- Stationary distribution: densities $f_0(x), f_1(x)$ on $(0, \infty)$ plus point mass $\pi_1^{(0)} = \Pr(n = 1, x = 0)$

Stability is not automatic

For $B < \infty$ the state space is compact $\Rightarrow$ stationary distribution always exists.

For $B = \infty$ the battery can drift to $+\infty$: stationarity requires

$$\underbrace{\frac{p_0}{\lambda}}_{\text{mean energy added per cycle}} < \underbrace{\frac{c}{\mu p_2}}_{\text{mean energy removed per cycle}}$$

Stability Condition

Theorem (Stability, $B = \infty$)

Define the ***key eigenvalue***

$$\theta_1 := \frac{\mu p_2}{c} - \frac{\lambda}{p_0}.$$

A proper stationary distribution exists if and only if

$$\boxed{\theta_1 < 0 \quad \Longleftrightarrow \quad \lambda > \lambda_c := \frac{\mu p_2 p_0}{c}}$$

Regime interpretation:

- $\theta_1 > 0$: battery drifts $\to \infty$ (station too lightly loaded)
- $\theta_1 < 0$: stable, battery bounded
- $\theta_1 = 0$: critical, [illegible]

Physical meaning:

The station must be *busy enough* to drain what the grid charges into the battery. λ_c balances mean energy input and output per cycle.

Proof of Stability: Finding the Lyapunov Function

The generator of $(n(t), x(t))$ acts on $V(n, x)$ as:

$$\mathcal{L}V(n, x) = r(n, x)\,\partial_x V + \lambda(n, x)\,[V(1 - n, x) - V(n, x)]$$

First attempt: $V(n, x) = x$.

$$\mathcal{L}V(0, x) = p_0 > 0, \qquad \mathcal{L}V(1, x) = -c < 0.$$

Fails: drift is positive in state $n = 0$ for all x, so no compact set can rescue it.

Second attempt: $V(n, x) = x + \alpha n$, $\alpha > 0$.

$$\mathcal{L}V(0, x) = p_0 + \lambda\alpha > 0, \qquad \mathcal{L}V(1, x) = -c - \mu p_2 \alpha < 0.$$

Still fails: no α makes both simultaneously negative.

The right idea

Use $V(n, x) = x + h_n$ where h_n depends only on n and is chosen so that $\mathcal{L}V$ is the **same constant** in both states. This requires solving one linear equation for the difference $h_0 - h_1$.

Proof of Stability: The Corrected Lyapunov Function

Ansatz: $V(n,x) = x + h_n$, h_n constant in x. Compute:

$$\mathcal{L}V(0,x) = p_0 + \lambda(h_1 - h_0) =: -\varepsilon$$
$$\mathcal{L}V(1,x) = -c + \mu p_2(h_0 - h_1) =: -\varepsilon$$

Setting $d := h_0 - h_1$ and equating the two expressions:

$$p_0 - \lambda d = -c + \mu p_2 d \quad \Rightarrow \quad \boxed{d = h_0 - h_1 = \frac{p_0 + c}{\lambda + \mu p_2}}$$

Substituting back gives the common drift value:

$$-\varepsilon = p_0 - \lambda d = \frac{p_0(\lambda + \mu p_2) - \lambda(p_0 + c)}{\lambda + \mu p_2} = \frac{p_0 \mu p_2 - \lambda c}{\lambda + \mu p_2}$$

Theorem (Foster–Lyapunov Certificate)

Take $h_0 = (p_0 + c)/(\lambda + \mu p_2)$, $h_1 = 0$. Then for all $x > 0$:

$$p_0 \mu p_2 - \lambda c$$

Balance Equations

Interior $(x > 0)$:

$$p_0 f_0'(x) = -\lambda f_0(x) + \mu p_2 f_1(x) \tag{E0}$$

$$-c f_1'(x) = \quad \lambda f_0(x) - \mu p_2 f_1(x) \tag{E1}$$

Boundary at $x = 0$:

$$p_0 f_0(0^+) = \mu p_1 \pi_1^{(0)} = c f_1(0^+)$$

Matrix form

Writing $\mathbf{f}' = A\mathbf{f}$ with $A = \mathbf{D}^{-1}Q$, $\mathbf{D} = \mathrm{diag}(p_0, -c)$, Q the M/M/1/1 generator:

$$A = \begin{pmatrix} -\lambda/p_0 & \mu p_2/p_0 \\ \lambda/c & -\mu p_2/c \end{pmatrix}$$

Characteristic equation: $\theta(\theta - \theta_1) = 0$, eigenvalues 0 and θ_1.

Solving the Balance Equations ($B = \infty$, $\theta_1 < 0$)

General solution:

$$\begin{pmatrix} f_0(x) \\ f_1(x) \end{pmatrix} = \alpha_0 \begin{pmatrix} \mu p_2 \\ \lambda \end{pmatrix} + \alpha_1 \begin{pmatrix} c \\ p_0 \end{pmatrix} e^{\theta_1 x}$$

Normalizability ($B = \infty$, $\theta_1 < 0$): requires $e^{\theta_1 x} \to 0$, but the α_0 mode is a *constant* — not integrable on $[0, \infty)$.

$$\Rightarrow \alpha_0 = 0$$

Theorem (Stationary Distribution, $B = \infty$)

Under $\theta_1 < 0$, with normalization constant $K_\infty = \dfrac{p_0 + c}{|\theta_1|} + \dfrac{p_0 c}{\mu p_1}$:

$$f_0(x) = \frac{c}{K_\infty} e^{\theta_1 x}, \quad f_1(x) = \frac{p_0}{K_\infty} e^{\theta_1 x}, \quad \pi_1^{(0)} = \frac{p_0 c}{\mu p_1 K_\infty}$$

Proportionality Property

Corollary (Decoupling in the interior)

For all $x > 0$:

$$\frac{f_1(x)}{f_0(x)} = \frac{p_0}{c} = const$$

The conditional probability of a car being present, given battery level $x > 0$, is constant and independent of x:

$$\Pr(n = 1 \mid x) = \frac{p_0}{p_0 + c}, \qquad \Pr(n = 0 \mid x) = \frac{c}{p_0 + c}$$

Interpretation

The battery level and the charger occupancy are **decoupled in the interior**: knowing x gives no information about whether a car is being served. The full distribution is characterized by the single marginal density

What Changes for Finite Battery

- Upper boundary $x = B$ is now a **reflecting boundary** when $n = 0$
- A new point mass arises: $\pi_0^{(B)} = \Pr(n = 0,\ x = B)$
- Stationary distribution **always exists** (compact state space)

Additional boundary conditions at $x = B$:

$$\lambda\, \pi_0^{(B)} = p_0\, f_0(B^-), \qquad c\, f_1(B^-) = \lambda\, \pi_0^{(B)}$$

These say: probability flux arriving at $x = B$ (from the p_0 drift) feeds $\pi_0^{(B)}$, which is drained by arrivals that immediately start discharging the battery.

Interior equations

Identical to the $B = \infty$ case:

$$p_0\, f_0'(x) = -\lambda\, f_0(x) + \mu p_2\, f_1(x), \qquad -c\, f_1'(x) = \lambda\, f_0(x) - \mu p_2\, f_1(x)$$

Same general solution; boundary conditions now determine α_0 and α_1.

Solving for the Stationary Distribution ($B < \infty$)

Boundary at $x = B$: flux condition $c\,f_1(B) = p_0\,f_0(B)$ gives

$$\alpha_0(c\lambda - p_0\mu p_2) = 0$$

Since $\theta_1 \neq 0 \Rightarrow c\lambda \neq p_0\mu p_2$, we again get $\alpha_0 = 0$.

Theorem (Stationary Distribution, $B < \infty$, $\theta_1 \neq 0$)

$$f_0(x) = \frac{c}{K}e^{\theta_1 x}, \quad f_1(x) = \frac{p_0}{K}e^{\theta_1 x}, \qquad x \in (0, B)$$

$$\pi_1^{(0)} = \frac{p_0 c}{\mu p_1 K}, \qquad \pi_0^{(B)} = \frac{p_0 c}{\lambda K}e^{\theta_1 B}$$

where

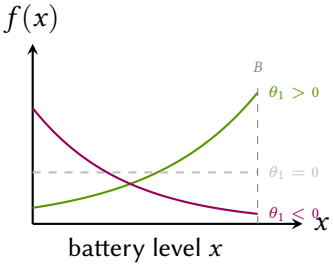
$$K = \frac{p_0 + c}{\theta_1}\left(e^{\theta_1 B} - 1\right) + \frac{p_0 c}{\mu p_1} + \frac{p_0 c}{\lambda}e^{\theta_1 B}$$

Remark

Regime Diagram

Sign of θ_1 governs the regime:

- $\theta_1 > 0$: density *increasing* toward B
  battery tends to be **full**
  $\Rightarrow$ fast charging almost always available
- $\theta_1 = 0$: uniform density
  critical balance
- $\theta_1 < 0$: density *decreasing* toward 0
  battery tends to be **empty**
  $\Rightarrow$ slow charging prevalent; B is the key lever



Blocking Probability

By **PASTA** (Poisson arrivals see time averages), arriving cars see the stationary distribution: $P_B = \Pr(n = 1)$.

Theorem (Blocking Probability)

$$P_B = \frac{p_0}{K}\left[\frac{e^{\theta_1 B} - 1}{\theta_1} + \frac{c}{\mu p_1}\right]$$

Limits:

$$P_B \xrightarrow{B \to 0} \frac{\lambda}{\lambda + \mu p_1} \qquad \textit{(M/M/1/1 with slow rate)}$$

$$P_B \xrightarrow{B \to \infty} \frac{\lambda}{\lambda + \mu p_2} \qquad \textit{(M/M/1/1 with fast rate)}$$

- Every admitted car finds $x > 0$ (since $\pi_0^{(0)} = 0$): **fast charging begins immediately**

Battery Depletion and Charging Time

Theorem (Depletion Probability)

An admitted car with demand y depletes the battery iff $y > p_2 x/c$. Averaging over the battery level seen at admission:

$$P_D(y) = \begin{cases} \dfrac{A}{\theta_1}\left(e^{\theta_1 c y/p_2} - 1\right) & y < p_2 B/c \\ 1 & y \geq p_2 B/c \end{cases}$$

where $A = c/[(1 - P_B)K]$ normalizes the admitted battery density.

Theorem (Charging Time Decomposition)

Given admission at battery level x with demand y:

$$T(x, y) = \frac{y}{p_2} + \left(\frac{1}{p_1} - \frac{1}{p_2}\right) \max\left(y - \frac{p_2 x}{c}, 0\right)$$

Mean Charging Efficiency

Theorem (Mean Charging Efficiency)

Let $\gamma := \theta_1 - \mu p_2/c = -\lambda/p_0 < 0$ *(always). Averaging* $T(x, y)/y$ *over* $Y \sim \mathrm{Exp}(\mu)$ *and the admitted battery density:*

$$\frac{E\left[[]\ T\right]}{E\left[[]\ Y\right]} = \frac{1}{p_2} + \left(\frac{1}{p_1} - \frac{1}{p_2}\right)\left[\frac{A(e^{\gamma B} - 1)}{\gamma} + g_B\, e^{-\mu p_2 B/c}\right]$$

where $g_B = \pi_0^{(B)}/(1 - P_B)$ *is the point mass at* $x = B$ *in the admitted distribution.*

Universal convergence

Since $\gamma = -\lambda/p_0 < 0$ **always**, regardless of other parameters:

$$\frac{E\left[[]\ T\right]}{E\left[[]\ Y\right]} \xrightarrow{B\to\infty} \frac{1}{p_2}$$

Enlarging the battery *always* improves efficiency, converging to the ideal fast-charge rate

System-Level Metrics

Theorem (Mode Fractions, Mean Battery, Mean Grid Power)

$$\rho_{\text{slow}} = \frac{p_0 c}{\mu p_1 K}, \quad \rho_{\text{fast}} = \frac{p_0 (e^{\theta_1 B} - 1)}{\theta_1 K}, \quad \rho_{\text{idle}} = \frac{c(e^{\theta_1 B} - 1)}{\theta_1 K} + \frac{p_0 c}{\lambda K} e^{\theta_1 B}$$

with $\rho_{\text{slow}} + \rho_{\text{fast}} + \rho_{\text{idle}} = 1$ *and* $\rho_{\text{slow}} + \rho_{\text{fast}} = P_B$.
Mean battery level:

$$E\left[[]\, x\right] = \frac{1}{K}\left[(p_0 + c)\left(\frac{Be^{\theta_1 B}}{\theta_1} - \frac{e^{\theta_1 B} - 1}{\theta_1^2}\right) + \frac{p_0 c B}{\lambda} e^{\theta_1 B}\right]$$

Mean grid power ($\in [p_0, p_1]$)*:*

$$E\left[[]\, P_{\text{grid}}\right] = p_1 - (p_1 - p_0)\, \rho_{\text{idle}}$$

- Larger $B \Rightarrow$ more idle time $\Rightarrow$ lower mean grid draw
- $\rho_{\text{slow}} \to 0$ as $B \to \infty$: slow charging disappears

Summary of Metrics

Metric	**Formula**	**Limit as $B \to \infty$**
Blocking P_B	$\frac{p_0}{K}\left[\frac{e^{\theta_1 B}-1}{\theta_1} + \frac{c}{\mu p_1}\right]$	$\frac{\lambda}{\lambda+\mu p_2}$
Depletion $P_D(y)$	$\frac{A}{\theta_1}(e^{\theta_1 cy/p_2} - 1)$	$1 - e^{\theta_1 cy/p_2}$
Mean efficiency	$\frac{1}{p_2} + (\frac{1}{p_1} - \frac{1}{p_2})[\cdots]$	$\frac{1}{p_2}$
Mean battery $E[[]\, x]$	closed form above	∞ (grows $\propto B$)
Mean grid power	$p_1 - (p_1 - p_0)\rho_{\text{idle}}$	$p_1 - (p_1 - p_0)(1 - P_B)$
ρ_{fast}	$\frac{p_0(e^{\theta_1 B}-1)}{\theta_1 K}$	$P_B - \rho_{\text{slow}}$

- All metrics are **explicit** functions of $(p_0, p_1, p_2, \mu, \lambda, B)$
- Single-exponential structure of f_0, f_1 drives all formulas

Why Exponentiality Was Special

For $Y \sim \mathrm{Exp}(\mu)$:

- Residual demand at any time $\sim \mathrm{Exp}(\mu)$ (memoryless)
- Departure rate is a function of (n, x) alone
- Process (n, x) is Markov $\Rightarrow$ ODE system in x

For general $Y \sim F$:

- Residual demand depends on **how long the car has been served** and **the charging rate history**
- Departure rate depends on the residual demand, not just on (n, x)
- (n, x) is **not** Markov

Augmented state

Let $r(t)$ = residual energy demand of the car currently in service (0 if $n = 0$). Then $(n(t), x(t), r(t))$ is Markov and defines a PDMP on

$$E = \{0\} \times [0, B] \times \{0\} \ \cup \ \{1\} \times [0, B] \times [0, \infty)$$

Augmented PDMP

Flow:

$$\begin{pmatrix}\dot{x}\\ \dot{r}\end{pmatrix} = \begin{cases} \begin{pmatrix}+p_0\\ 0\end{pmatrix} & n=0 \\ \begin{pmatrix}-c\\ -p_2\end{pmatrix} & n=1,\ x>0 \\ \begin{pmatrix}0\\ -p_1\end{pmatrix} & n=1,\ x=0 \end{cases}$$

Jumps:

Arrivals (rate λ, when $n=0$):

$$(0,x,0) \ \to\ (1,x,r') \quad r' \sim f_Y$$

Departures (when r hits 0):

$$(1,x,0) \ \to\ (0,x,0)$$

These are **forced jumps** at the boundary $r=0$, in the sense of Davis (1984).

Two regions of interest

- **Region I:** $n=1$, $x>0$, $r>0$ — fast charging; drift $(\dot{x},\dot{r}) = (-c,-p_2)$
- **Region II:** $n=1$, $x=0$, $r>0$ — slow charging; drift $(0,-p_1)$

Main Theorem: General Demands (I)

Theorem (Characteristic Solution in Region I)

Let $f_0(x)$ be the stationary density for $n = 0$, $x > 0$. In Region I ($n = 1$, $x > 0$, $r > 0$), the joint density $f_1(x, r)$ satisfies the transport PDE with source:

$$c\,\partial_x f_1 + p_2\,\partial_r f_1 = \lambda\, f_0(x)\, f_Y(r)$$

The characteristics are lines $p_2 x - cr = \text{const}$; integrating along them from the entry point of each trajectory:

$$\boxed{f_1(x, r) = \frac{\lambda}{p_2} \int_r^{\infty} f_0\left(x + \frac{c(R - r)}{p_2}\right) f_Y(R)\, dR}$$

Interpretation: the density at (x, r) sums contributions from all cars that arrived at battery level $x + c(R - r)/p_2$ with demand $R > r$ and have since been partially served.

Main Theorem: General Demands (II)

Theorem (Laplace Transform of f_0)

Let $M_Y(t) = E[[]\, e^{tY}]$ be the MGF of Y. Using the departure flux $p_2 f_1(x, 0^+)$ from the characteristic solution to close the f_0 balance equation $p_0 f_0' = -\lambda f_0 + p_2 f_1(\cdot, 0^+)$, and taking the Laplace transform in x:

$$\boxed{\hat{f}_0(s) = \frac{p_0\, f_0(0^+)}{p_0 s + \lambda\left[1 - M_Y\left(\frac{cs}{p_2}\right)\right]}}$$

where $f_0(0^+)$ is fixed by normalization, and the denominator $\psi(s) := p_0 s + \lambda[1 - M_Y(cs/p_2)]$ is the ***characteristic function*** *of the problem.*

- The poles of $\hat{f}_0$ (zeros of ψ) determine the exponential modes of f_0
- For $Y \sim \mathrm{Exp}(\mu)$: one pole $\Rightarrow$ single exponential
- For $\mathrm{Erlang}(n, \mu)$: n poles $\Rightarrow$ sum of n exponentials

Proof Sketch

Step 1 — Transport PDE with source. The balance for $f_1(x, r)$ in Region I (no sources from departures in the interior; arrivals inject mass at rate $\lambda f_0(x) f_Y(r)$):

$$c\,\partial_x f_1 + p_2\,\partial_r f_1 = \lambda\, f_0(x)\, f_Y(r)$$

Step 2 — Integrate along characteristics. The characteristic through (x, r) was seeded by a car arriving at battery level $X = x + c(R - r)/p_2$ with demand R. Since each trajectory passes through (x, r) exactly once at speed p_2 in the r-direction:

$$f_1(x, r) = \frac{\lambda}{p_2} \int_r^{\infty} f_0\left(x + \frac{c(R - r)}{p_2}\right) f_Y(R)\, dR$$

Step 3 — Close via the f_0 equation. The f_0 balance is: $p_0 f_0'(x) = -\lambda f_0(x) + p_2 f_1(x, 0^+)$. Substituting $f_1(x, 0^+)$ from Step 2 and taking the Laplace transform:

$$\hat{f}_0(s)\left[p_0 s + \lambda - \lambda\, M_Y\left(\frac{cs}{p_2}\right)\right] = p_0\, f_0(0^+) \qquad \square$$

Recovery of the Exponential Case

For $Y \sim \mathrm{Exp}(\mu)$: $M_Y(t) = \mu/(\mu - t)$, so

$$1 - M_Y\left(\frac{cs}{p_2}\right) = \frac{-cs/p_2}{\mu - cs/p_2}$$

Substituting into $\hat{f}_0(s) = p_0 f_0(0^+)/\psi(s)$ and simplifying:

$$\hat{f}_0(s) = \frac{p_0 f_0(0^+)}{s - \theta_1} \xrightarrow{\mathcal{L}^{-1}} f_0(x) = p_0 f_0(0^+)\, e^{\theta_1 x} \quad (\theta_1 < 0)$$

which is exactly the single-exponential solution derived by the ODE method. ✓

General F: poles of $\hat{f}_0$ determine the modes of f_0

Distribution F	$M_Y(cs/p_2)$	Modes of f_0
$\mathrm{Exp}(\mu)$	Möbius (rational, deg 1)	1 exponential
$\mathrm{Erlang}(n, \mu)$	Rational, deg n	n exponentials
Hyper-exp	Rational	Finite sum

Characteristic Equation and Zeros

Define the **characteristic function**:

$$\psi(s) := p_0\, s + \lambda\left[1 - M_Y\left(\frac{cs}{p_2}\right)\right]$$

- $\psi(0) = 0$ always (trivial zero, corresponds to normalization)
- $\psi'(0) = p_0 - \dfrac{\lambda c}{\mu p_2}$, so $\psi'(0) < 0 \iff \theta_1 < 0$
- For stability (finite $E\,[[]\,x]$): need $\psi'(0) < 0$, i.e. $\theta_1 < 0$ — same condition as exponential case!

Stability condition is distribution-free

The stability condition $\theta_1 < 0$ (equivalently $\lambda > \lambda_c = \mu p_2 p_0 / c$) depends on F **only through its mean** $1/\mu$. The same critical load λ_c governs stability for any demand distribution with mean $1/\mu$.

Performance Metrics for General F: Blocking

Theorem (Blocking Probability, General F, $B = \infty$)

By PASTA, $P_B = \Pr(n = 1)$ *and* $1 - P_B = \lambda \int_0^\infty f_0(x)\, dx = \lambda \hat{f}_0(0)$. *Since* $\psi(0) = 0$, *L'Hôpital gives:*

$$\hat{f}_0(0) = \frac{p_0 f_0(0^+)}{\psi'(0)}, \qquad \psi'(0) = p_0 - \frac{\lambda c}{\mu p_2}$$

Combined with the normalization $\int_0^\infty (f_0 + f_1) dx + \pi_1^{(0)} = 1$, *this yields:*

$$\boxed{P_B = \frac{\lambda c/(\mu p_2 p_0)}{1 + \lambda c/(\mu p_2 p_0) + \mu p_1\, \pi_1^{(0)}/(p_0 f_0(0^+))}}$$

For the special case where $\pi_1^{(0)} \to 0$ *(battery rarely empty, large B):*

$$P_B \approx \frac{\lambda/\lambda_c}{1 + \lambda/\lambda_c}$$

Mean Battery Level, General F

Theorem (Mean Battery Level, General F, $B = \infty$)

Since $\mathrm{E}_0[x] = -\hat{f}_0'(0)$, *differentiate* $\hat{f}_0(s) = p_0 f_0(0^+)/\psi(s)$ *and evaluate at* $s = 0$ *using L'Hôpital (since* $\hat{f}_0$ *has a simple pole at* $s = 0$*):*

$$\mathrm{E}_0[x] = \frac{p_0\, f_0(0^+)\, \psi''(0)}{2\, \psi'(0)^2}$$

Computing the second derivative of ψ *at zero:*

$$\psi''(0) = -\frac{\lambda c^2}{p_2^2}\, E\left[Y^2\right]$$

so that:

$$\boxed{\mathrm{E}_0[x] = \frac{\lambda c^2\, E\left[Y^2\right]}{2p_2^2} \cdot \frac{p_0\, f_0(0^+)}{\psi'(0)^2}}$$

Mean Charging Efficiency, General F

Theorem (Mean Charging Time, General F)

The mean service time decomposes as:

$$E[[]\,T] = \frac{E[[]\,Y]}{p_2} + \left(\frac{1}{p_1} - \frac{1}{p_2}\right) \int_0^\infty g(x_0) \int_{p_2 x_0/c}^\infty \bar{F}(y)\, dy\, dx_0$$

where $g(x_0)$ is the battery density seen at admission and $\bar{F}(y) = \Pr(Y > y)$ is the tail of F.

- Uses the **stop-loss identity**: $E[[]\max(Y - t, 0)] = \int_t^\infty \bar{F}(y)\, dy$
- For $Y \sim \mathrm{Exp}(\mu)$: $\bar{F}(y) = e^{-\mu y}$, integral is explicit $\Rightarrow$ recovers earlier formula
- For general F: $E[[]\,T]/E[[]\,Y]$ depends on the entire distribution through $\bar{F}$

Convergence to $1/p_2$ as $B \to \infty$

As $B \to \infty$, admitted customers increasingly find a full battery, $g(x_0) \to 0$ for small x_0, and the depletion integral vanishes:

$$E[T] \qquad 1$$

What Changes vs the Exponential Case

Feature	**Exponential** Y	**General** Y
State space	(n, x)	(n, x, r)
Departure mechanism	Poisson, rate $\delta(x)$	Deterministic hitting $r = 0$
Balance equations	ODE system in x	PDE system in (x, r)
Solution method	Eigenvalue decomposition	Characteristics + Laplace
$\hat{f}_0(s)$	$A/(s - \theta_1)$	$p_0 f_0(0^+)/\psi(s)$
$f_0(x)$	Single exponential	Sum of exponentials
Proportionality f_1/f_0	Constant in x	x-dependent
Stability condition	$\lambda > \lambda_c$	Same λ_c (mean only)
P_B	Explicit	Via $\psi'(0)$; depends on mean only
$E[x]$	Explicit	Via $\psi''(0)$; needs $E\left[Y^2\right]$
$E[T]/E[Y]$	Explicit	Via stop-loss transform of F

Summary and Open Problems

Summary:

- Modeled battery-buffered EV charging station as a PDMP on $\{0,1\} \times [0,B]$
- Exponential demands: exact closed-form stationary distribution and metrics
- Key structural result: proportionality $f_1/f_0 = p_0/c$ (charger occupancy independent of battery level)
- General demands: solvable via characteristics + Laplace transform; $\hat{f}_0(s) = p_0 f_0(0^+)/\psi(s)$
- Stability condition $\lambda > \lambda_c$ is **distribution-free** (depends only on mean demand)
- Metrics for general F involve $E\left[Y^2\right]$, stop-loss transform, MGF

Open problems:

- Multi-server extension ($s > 1$, shared grid)
- Finite B with general F: boundary conditions couple Regions I and II
- Waiting room (non-loss) systems
- Phase-type demands: matrix-analytic approach
- Optimal sizing of B, p_0, p_2 given cost and revenue